

SleepECGFusion: a cross-modal deep learning framework for automatic sleep stage classification using single-lead ECG

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ABSTRACT

Accurate and readily accessible sleep stage classification is crucial for diagnosing sleep disorders. However, conventional polysomnography (PSG) is cumbersome. Given the convenience of single-lead electrocardiogram (ECG) signals, we propose a novel cross-modal deep learning framework named SleepECGFusion. It aims to achieve high-precision automated sleep stage classification and apnea detection. SleepECGFusion ingeniously integrates multi-scale time-domain features extracted from raw ECG sequences and high-level spectral features derived from time-frequency (TF) maps for high-level spectral representation. The framework consists of a 1) temporal branch based on multiscale feature extraction (MSFE), convolutional attention module (CBAM) and bidirectional long- and short-term memory (BiLSTM) network and 2) time-frequency branch with EfficientNet-B0 as the backbone. We systematically evaluated the effect of input signal duration on model performance, thereby revealing that longer segments significantly enhance classification accuracy across all sleep stages. Furthermore, SleepECGFusion achieves state-of-the-art performance in ECG-based sleep stage classification tasks. To rigorously assess generalizability, we conducted cross-dataset performance benchmarking using SHHS1 for training and ISRUC-S3 for testing in a subject-independent manner, demonstrating robust performance across diverse populations. The framework also demonstrates strong generalization capabilities for the detection of sleep apnea via transfer learning and obtains comparable classification results between healthy individuals and sleep apnea patients. Overall, this work emphasizes the significant potential of single-lead ECG as a viable, non-invasive alternative to PSG. It lays the foundation for scalable and accessible sleep health monitoring and diagnosis.

1. Introduction

Sleep stage classification is of high significance for understanding sleep quality and diagnosing sleep-related diseases [1,2]. According to the International Classification of Sleep Disorders, sleep is mainly categorized into rapid-eye movement (REM) sleep and non-rapid eye movement (NREM) sleep. NREM sleep is further divided into N1, N2 and N3 stages [3]. Polysomnography (PSG) is a conventional method for sleep stage classification. It monitors sleep states by recording physiological signals including electroencephalogram (EEG), electrocardiogram (ECG), electrooculogram (EOG), and electromyogram (EMG) [4]. The analysis of these signals is affected significantly by generally expe-

rience and subjective state, and the interpretation results often exhibit subjective differences [5].

Considering that the manual analysis of PSG signals is both time-consuming and labor-intensive, many recent studies have leveraged deep learning techniques on PSG signals for sleep stage classification [6]. However, it is challenging for most existing approaches to effectively capture the interaction features between different modalities when processing multi-modal signals [7–10]. Additionally, these methods generally exhibit limited adaptability to downstream tasks [5]. Moreover, the majority of the current studies rely on multi-channel signals or EEG data for sleep stage classification [11,12]. Although such signals provide rich physiological information, the process of wearing

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associated equipment generally causes discomfort for subjects. In contrast, single-lead signals present a more convenient and comfortable alternative for sleep stage classification [13].

However, when utilizing single-lead ECG signals for sleep analysis, the information available to the model is generally limited [14,15]. Although convenient and comfortable, ECG signals primarily reflect cardiac activity and do not provide direct information regarding brainwave activity provided by EEG. This largely limits the potential of relying on ECG signals for classifying sleep stages. Based on the above analysis, we summarize certain major challenges currently confronting the field of automated sleep stage classification: 1) how to effectively utilize single-lead ECG information to overcome the limitations of a single modality and 2), most of the existing models are task specific, and their generalization capability is generally insufficient when being migrated from sleep stage classification tasks to other sleep-related diagnostic tasks.

As illustrated in Fig. 1, this study proposes a novel cross-modal fusion framework SleepECGFusion. It is designed to achieve high-accuracy automatic sleep stage classification based on single-lead ECG signals. The framework integrates raw ECG sequences and their corresponding time-frequency (TF) representations to extract complementary features from both temporal and spectral domains [16]. Specifically, the ECG branch incorporates a multi-scale feature extraction (MSFE) module to capture local and global temporal features across multiple resolutions [17,18]. This is followed by a convolutional block attention module (CBAM) to enhance the representation of salient information [19], and a bidirectional long short-term memory (Bi-LSTM) network to model long-range dependencies [20]. In parallel, the TF branch adopts EfficientNet-B0 as its backbone to effectively extract high-level semantic features from spectrogram images [21]. The modality-specific features from both branches are then aligned and combined into a unified representation. This unified representation is fed into a classification module to predict the sleep stages. This study aims to enhance the model performance under the conditions on inexpensive, wearable device; evaluate its feasibility without relying on PSG; and validate its transferability and adaptability to populations with sleep disorders.

Our work makes several contributions to the field of sleep stage classification. To address the current challenges in automated sleep stage classification, the sleep fusion framework proposed in this study (SleepECGFusion) effectively integrates the time-domain characteristics of raw ECG signals with the spectral characteristics of the TF representation. This enables the learning of multimodal complementary features and further demonstrates its strong generalization capability for detecting sleep apnea through transfer learning [22–24]. In particular, the model provides comparable classification results between healthy individuals and patients with sleep apnea, which fully verifies its robustness. Furthermore, we systematically evaluated the effect of different input durations on the performance of the sleep stage classification model. It was observed that longer input durations significantly improved the model across all sleep stages. This validates the critical role of long temporal information in capturing sleep stage features.

The primary contributions of this study are as follows:

- 1) We propose a decision-level fusion framework tailored for single-lead ECG signals, integrating multiscale temporal features from raw ECG sequences with spectral features from time-frequency representations. This dual-branch design enables independent optimization of complementary temporal and spectral information, enhancing the model's ability to capture ECG-specific patterns and significantly improving the accuracy of sleep stage classification.
- 2) Our approach achieves state-of-the-art performance in ECG-based sleep stage classification. To further evaluate the model's generalizability, we additionally performed cross-dataset, subject-independent validation (SHHS1 → ISRUC-S3), demonstrating robust performance across diverse populations. Furthermore, the framework exhibits strong transfer learning capability for the detection of sleep apnea, achieving comparable classification results for healthy

individuals and patients with sleep apnea. These comprehensive evaluations verify the model's robustness and practical applicability for real-world sleep monitoring scenarios.

- 3) We systematically evaluated the effect of different input durations on the performance of the sleep stage classification model. It was observed that longer input durations significantly improved the accuracy of the model classification across all sleep stages, particularly in N1, N3 and REM. This validates the hypothesis that long time series information helps capture the characteristics of the sleep stage.

The remainder of this paper provides a comprehensive overview of our work. A detailed review of related work is presented in Section 2, and Section 3 describes the theoretical foundations and methodological framework of SleepECGFusion. Section 4 presents detailed experimental methods. Section 5 presents and discusses the overall experimental results and Section 6 presents a detailed analysis of the significance and limitations of our results. Section 7 describes the prospects for the application of SleepECGFusion and directions for future improvements.

2. Related works

Sleep stage classification is the process of dividing sleep into stages. It serves as the basis for assessing sleep quality and diagnosing various sleep-related disorders [4]. Conventionally, PSG involves the recording of multiple physiological signals. It is the gold standard for sleep stage classification [6]. However, the complexity, high cost, invasiveness, and need for a specialized environment of a multisensor setup inherently limit its popularity and applicability for long-term monitoring. In addition, the manual interpretation of PSG signals by experts is time-consuming and laborious, and is susceptible to inter-reader variability owing to subjective factors [25].

To overcome these limitations, significant research efforts have been dedicated to automating sleep stage classification. Early attempts generally relied on rule-based systems or conventional machine learning methods. These require extensive feature engineering from raw physiological signals [26–28]. With the advent of deep learning, automated feature extraction capabilities have propelled the field forward. This has resulted in the construction of various end-to-end models based on CNNs, RNNs, and LSTMs for multi-channel PSG data [28,29].

Recent advances in multi-modal deep learning have shown remarkable promise for automatic sleep staging. For example, Phan et al. [30] proposed XSleepNet using EEG/EOG/EMG with adaptive gradient blending, while Huttunen et al. [31] explored multisignal combinations (PPG, SpO2, nasal pressure) for simultaneous sleep staging and detection of respiratory events. Although these approaches have shown impressive performance, many still rely on multichannel PSG or multiple physiological signals, particularly EEG, which pose significant challenges regarding subject comfort, setup complexity, and practicality for long-term home-based monitoring. Single-lead ECG offers a more accessible and comfortable alternative, but remains underexplored for comprehensive sleep staging.

Given the limitations of conventional PSG and multichannel methods, research is increasingly focusing on classification of sleep stages using single-lead physiological signals that are inherently non-invasive and convenient to capture through wearable devices [32]. Researchers have evaluated a variety of single-channel modalities, including single-lead EEG, photoplethysmograph (PPG), and respiratory signals. Each of these can provide unique information on sleep physiology [33–35]. However, these signals display unique challenges such as the limited direct brain activity information and discomfort while wearing a single-lead EEG, sensitivity of PPGs to motion artifacts, or indirect reflection of breathing on sleep stages.

Single-lead ECG offers a practical alternative for sleep monitoring due to its physiological informativeness. ECG spectral characteristics reflect sympathetic and vagal nerve regulation [36], with muscle sympathetic nerve activity (MSNA) varying across sleep stages and modulat-

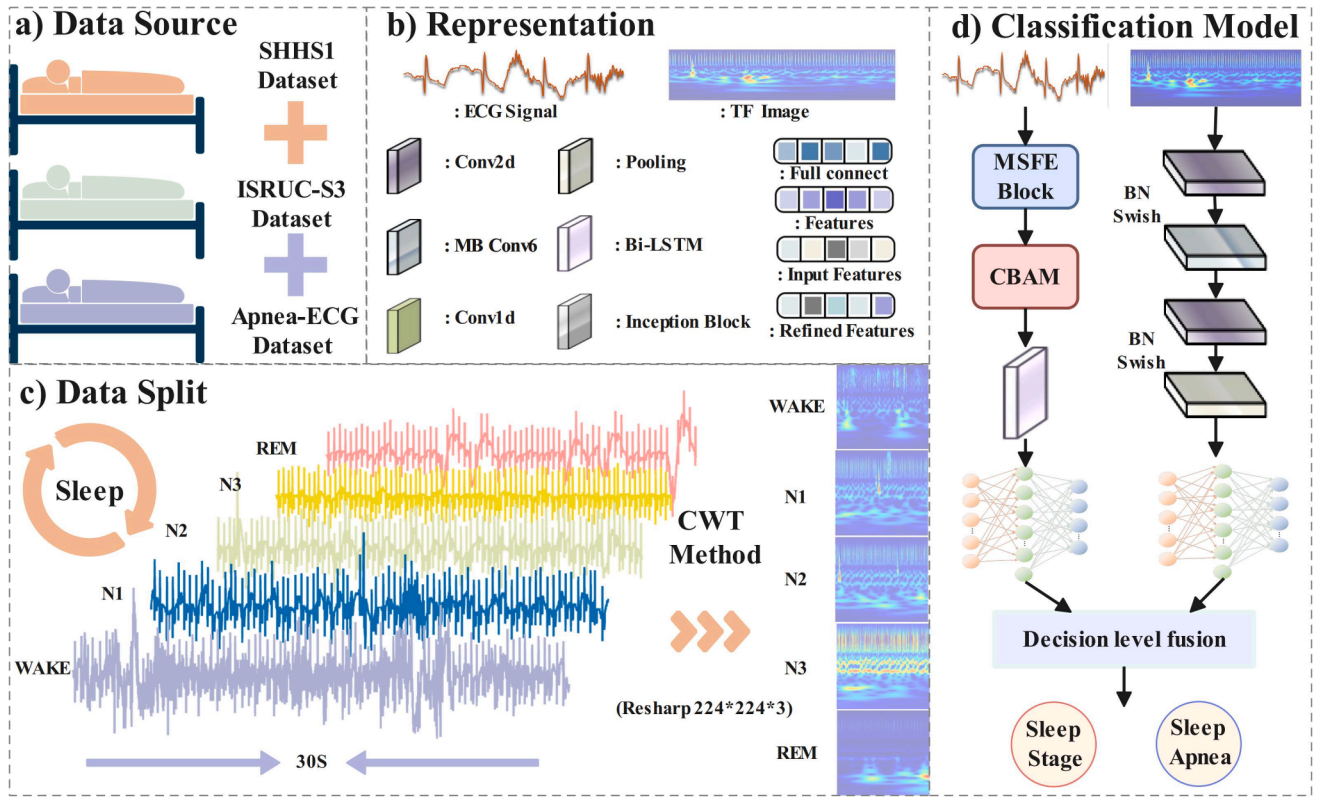


Fig. 1. The overall workflow of the SleepECGFusion framework. This framework consists of four main components: (a) Data sources: Utilizing three ECG datasets. (b) Characterization: Describing the components required by the framework using different elements. (c) Data Segmentation: 30-second intervals are annotated by sleep stage, with CWT applied to generate time-frequency images. (d) Classification Model: A dual-branch architecture integrates MSFE, CBAM, and BiLSTM for temporal modeling. EfficientNet-B0 processes the time-frequency images. Features from both branches undergo decision-level fusion to generate sleep stage classification results. Transfer learning is further applied to produce predictions for sleep apnea detection.

ing autonomic discharge [37]. This enables non-invasive assessment of sleep regulation and apnea detection. Importantly, ECG provides complementary information in time and frequency domains. Time-domain features capture morphological patterns and R-R interval dynamics, while time-frequency representations reveal spectral components and periodic patterns not readily apparent in temporal analysis [38]. Dual-domain approaches have been shown to significantly enhance ECG interpretation [39], motivating our framework that integrates both representations.

Initial studies employing single-lead ECG for sleep stage classification primarily utilized conventional machine learning techniques. These approaches typically involved the manual extraction of various time-domain, frequency-domain, and non-linear features derived from heart rate variability (HRV) and RR intervals [40,41]. For example, models such as support vector machines (SVM) [42], random forests [43], and hidden Markov models [44] were trained on these handcrafted features to classify sleep stages. Although these methods demonstrate the feasibility of ECG-based staging, they are inherently limited by the time-consuming and laborious characteristics of manual feature engineering and their incapability to fully capture complex, latent patterns within the raw physiological signals. Recent work has highlighted the importance of sequential dependencies in sleep stage classification. The language model-inspired approach improves prediction by modeling stage transitions, particularly for underrepresented stages like N1. This work advances the field through improved signal-level feature fusion, with potential benefits from future integration with sequence models [45].

More recently, deep learning methods have gained traction for ECG-based sleep stage classification. These provide automated feature

extraction from raw signals [46,47]. However, several challenges persist in these single-lead ECG deep learning approaches. First, relying only on a single modality (including ECG) can inherently limit the information available for fine-grained sleep stage differentiation. This is because ECG primarily reflects cardiac activity and does not yield the direct brain electrical signals provided by EEG [47]. This single-modality constraint generally causes difficulty in detecting the subtle differences between sleep stages. Second, notwithstanding automatic feature extraction, it can be challenging for current methods to effectively identify and focus on the most critical information within long time-series data. Herein, key features may be obscured by subsequent information, thereby potentially impacting the accuracy [11]. To overcome the limitations of short-time window modeling, Choi et al. proposed an end-to-end Transformer architecture that models full-night ECG sequences without explicit segmentation, demonstrating superior performance in handling sleep stage transitions [48]. Recent advances in temporal contrastive learning, such as temporal proximity contrastive learning (TP-CL) [49], further address temporal variability by learning representations robust to context variations while preserving event-related discriminative features. These approaches reveal the potential of long-term context modeling and temporal relationship learning to enhance sleep staging accuracy, offering promising directions for future research. Finally, the problem of class imbalance—where certain sleep stages are underrepresented significantly in datasets—remains a prevalent challenge, as it affects model robustness and generalization [50]. Addressing these limitations necessitates the development of advanced multi-modal integration strategies and robust loss functions capable of handling data imbalance and enhancing feature representation across diverse populations.

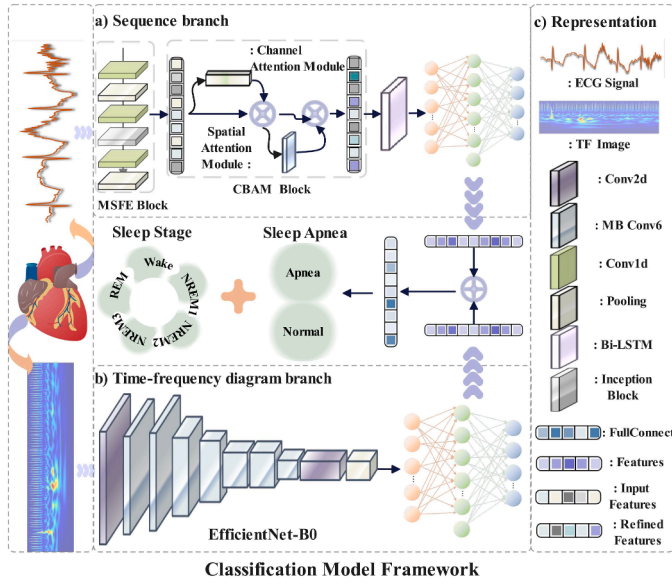


Fig. 2. The proposed SleepECGFusion framework for automatic sleep staging and apnea detection using single-lead ECG. The framework consists of two parallel branches: (a) Sequence branch processes raw ECG time-series through MSFE, CBAM, and Bidirectional BiLSTM to capture temporal dynamics; (b) Time-frequency diagram branch converts ECG signals into CWT spectrograms and extracts high-level spectral features using EfficientNet-B0. (c) Representation illustrates the multimodal inputs and the detailed architecture components. The framework outputs sleep stage classifications and supports transfer learning for apnea detection.

3. Method

3.1. Continuous wavelet transform

To eliminate amplitude-related effects, the single-lead ECG signals were first normalized to the interval $[0, 1]$. Then, the Morse wavelet coefficient matrix (size 87×3000) was calculated for each 30-second period using the continuous wavelet transform (CWT) [16]. Here, '87' and '3000' denote the frequency and time step of the signal corresponding to each column of the coefficient matrix, respectively. Each element of the matrix represents the energy or amplitude of the signal at a particular scale and time step. In this study, we used the cwt function in MATLAB R2024b to compute this matrix. The scalar plot was defined by plotting the absolute values of the wavelet coefficients in the time-frequency plane. Following previous studies [51,52], we selected the color mapping named "Jet" to map the wavelet coefficient matrix to the corresponding RGB color values. Thus, we obtained the original RGB scalar map. Finally, the image was resized to $224 \times 224 \times 3$ (the process is shown in Fig. 3) to prepare it for subsequent processing.

$$Q(t, \alpha) = \text{CWT}\{q(t)\} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{\alpha}} x(\tau) \psi\left(\frac{\tau-t}{\alpha}\right) d\tau \quad (1)$$

where $Q(t, \alpha)$ is the matrix of wavelet coefficients, $q(t)$ is the 30 s ECG, $\psi(t)$ is the mother wavelet, t is the scale parameter, and α is the translation parameter.

In addition, Fig. 4 counts the time-frequency plots for different periods in the sleep stage classification task and the time-frequency plots for normal versus onset conditions in the sleep apnea detection task. To verify the validity of the CWT method, subsequent experiments were performed using Hilbert-Huang transform (HHT), Gramian angular difference field (GADF), and Gramian angular summation field (GASF) methods for comparative experiments [53–55].

3.2. Sequence branch

In the cross-modal fusion framework proposed in this study, the temporal branch aims to extract rich time-domain feature information from the raw single-lead ECG signals to adequately model the dynamic evolution of the central electrical activity during sleep [56]. The overall description is placed in the upper half of Fig. 2 (a). The entire branch consists of three key modules in sequence: MSFE [17], CBAM [19], and BiLSTM [20]. Through autonomous sequence modeling, the temporal branch captures the time-dependent characteristics of R-R intervals and the morphological transitions between sleep stages [38]. These modules are described in detail below. First, the raw input of the ECG signal is a one-dimensional time series $x = \{x_1, x_2, \dots, x_T\}$, $x \in \mathbb{R}^{T \times 1}$. Here, T denotes the time duration of the sample (number of sampling points under the 30/60/90 second window). To enhance the model's capability to perceive multi-granularity temporal variations, we designed parallel multi-scale convolutional paths. Each of these uses a different size of convolutional kernel to extract local and global feature responses. Each convolutional path contains convolution, batch normalization (Batch-Norm) and nonlinear activation (ReLU) operations. Its i -th scale feature extraction can be expressed as:

$$f_i = \text{ReLU}\left(\text{BN}\left(\text{Conv1D}_{k_i}(x)\right)\right) \quad (2)$$

where k represents different convolutional kernels that splice features at all scales along the channel dimension to form a unified multiscale representation:

$$f_{\text{MSFE}} = \text{Concat}(f_1, f_2, f_3) \in \mathbb{R}^{T \times C} \quad (3)$$

where C represents the number of channels after splicing.

Next, the CBAM attention mechanism was introduced to enhance the temporal branching attention to critical physiological feature regions. CBAM first models the importance of each channel through the channel attention module. Specifically, the channel attention weights $M_c \in \mathbb{R}^{1 \times C}$ are obtained by global average pooling with maximum pooling. These are then fed into the shared multilayer perceptual machine (MLP) and activated by sigmoid. Subsequently, the significant channel responses are augmented by multiplication operations:

$$f_c = M_c \odot f_{\text{MSFE}} \quad (4)$$

Based on this, the temporal attention weight $M_s \in \mathbb{R}^{T \times 1}$ is then extracted by a convolution operation on the temporal dimension and further adjusted to represent the sequence at each time point:

$$f_{\text{CBAM}} = M_s \odot f_c \quad (5)$$

To efficiently capture the long-term dependencies of ECG sequences during the sleep cycle, we further introduced BiLSTM to model the attention-enhanced features. BiLSTM is capable of modeling both forward and backward sequence information. Its output is defined as:

$$f_{\text{BiLSTM}} = \text{BiLSTM}(f_{\text{CBAM}}) \in \mathbb{R}^{T \times H} \quad (6)$$

where H denotes the hidden state dimension of the BiLSTM. To obtain a fixed-length global temporal representation, an average pooling operation of the time dimension is finally applied to compress the states of all the time points to obtain a uniform representation vector:

$$f_{\text{temporal}} = \frac{1}{T} \sum_{t=1}^T f_{\text{BiLSTM}}(t) \quad (7)$$

This representation vector integrates multi-scale local features, attention-enhanced discriminative information, and long-range dependency structures. As the final output of the temporal branch, it is subsequently aligned and fused with the output features of the frequency-domain branch in the shared space to jointly accomplish the classification prediction of sleep stage classification.

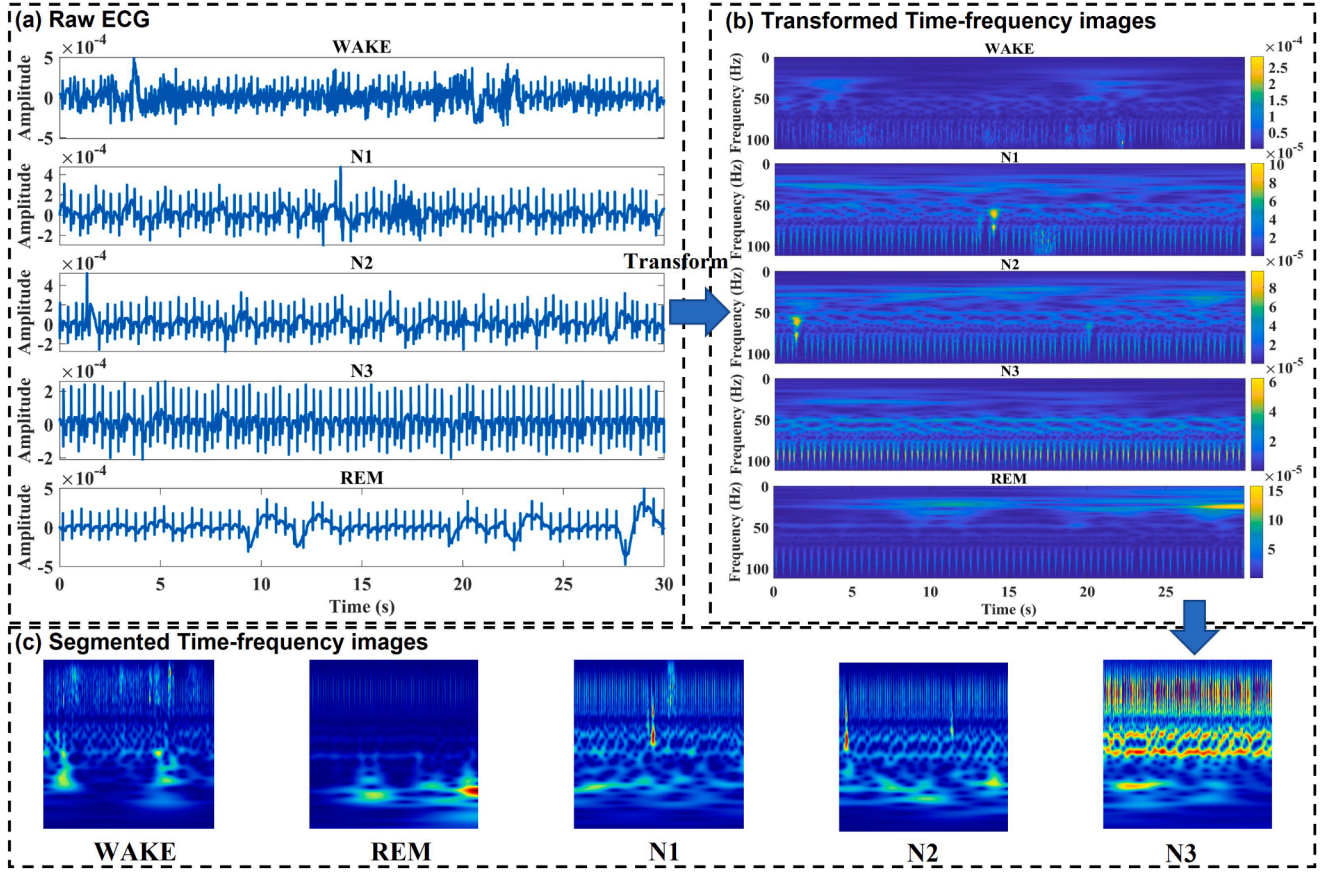


Fig. 3. Data processing pipeline of SleepECGFusion. (a) Raw single-lead ECG signals for different sleep stages. (b) Time-frequency representations obtained via Continuous Wavelet Transform. (c) Segmented time-frequency images for each sleep stage, serving as input to the spectral branch.

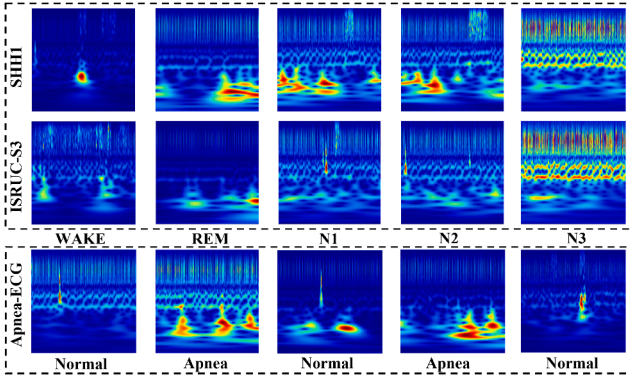


Fig. 4. Examples of typical time-frequency images, with rows corresponding to each of the three main datasets. The ECG TF images from the SHHS1 and ISRUC-S3 datasets were used to differentiate between the five different sleep stages, and TF images from the Apnea-ECG dataset focused on differentiating between the characteristics of normal and apnea.

3.3. Time-frequency diagram branch

The time-frequency branch extracts spectral features using CWT and EfficientNet-B0 to quantify autonomic nervous system balance and periodic patterns in the frequency domain [39]. After completing the continuous wavelet transform described in Section 3.1, the resulting color time-frequency map was scaled to an RGB image with a size of $224 \times 224 \times 3$ as the input to the time-frequency map branch. To effectively model the time-frequency patterns in this image, a lightweight convolutional neural network named EfficientNet-B0 was used as the

feature extraction backbone network in this study. This is shown in the lower part of Fig. 2 (b). This network introduces a compound scaling strategy in its structural design. It uniformly scales the depth, width, and input resolution of the network. Thus, it obtains a better representation capability while maintaining a lower computational complexity. The core building block of EfficientNet-B0 is the mobile backward bottleneck structure (MBConv). It incorporates depthwise separable convolution and residual connectivity mechanisms to effectively enhance the network's capability to model the local and global features of an image [21]. The input image $I \in \mathbb{R}^{224 \times 224 \times 3}$ is transformed gradually into a higher dimensional and semantically richer feature map $F \in \mathbb{R}^{H \times W \times C}$ after several stages of convolutional processing. Here H , W , and C denote the height, width, and number of channels of the output feature map, respectively. To compress the spatial dimensions while retaining the most representative global information, the final output feature map is compressed using a global average pooling (GAP) layer, yielding a fixed-length high-dimensional feature vector:

$$f_{\text{spec}} = \text{GAP}(\text{EfficientNetB0}(I)) \quad (8)$$

where $f_{\text{spec}} \in \mathbb{R}^C$ denotes the global time-frequency features extracted from the time-frequency map, which exhibit good discriminative capability and robustness. Unlike conventional one-dimensional convolutional features, the time-frequency branch provides a more intuitive and information-dense two-dimensional representation. This enables the model to effectively capture cross-scale and cross-frequency structural patterns in the ECG signal.

The features output from this branch are integrated with the time-domain multi-scale convolutional features extracted from the temporal branch in the subsequent fusion module, thereby realizing the complementary enhancement of time-domain and time-frequency domain

information. Through this cross-modal feature fusion strategy, the model identifies anomalous patterns within a broader sensory field and a richer feature space, which improves the overall identification performance and generalization capability. The cross-modal fusion strategy is elaborated in the next section.

3.4. Cross-modal fusion strategies

SleepECGFusion integrates temporal and spectral ECG representations to capture complementary autonomic information. Time-domain features reflect sequential autonomic patterns, including R-R variability and morphological changes [36], while time-frequency features quantify autonomic balance through HRV power distribution (VLF/LF/HF bands) and respiratory modulation [37]. These representations are orthogonal: temporal emphasizes sequence dynamics, while spectral emphasizes frequency content and periodicity [39]. We employ decision-level fusion to enable (1) independent optimization of each modality-specific architecture, (2) interpretability through separate predictions.

Specifically, let the temporal branch output a vector of category scores $z_{\text{temporal}} \in \mathbb{R}^K$ after feature extraction and classifier. The classification probability of the branch is obtained by the softmax function:

$$p_{\text{temporal}} = \text{Softmax}(z_{\text{temporal}}) = \frac{\exp(z_{\text{temporal}})}{\sum_{k=1}^K \exp(z_{\text{temporal}}^{(k)})} \quad (9)$$

Similarly, the time-frequency graph branch outputs the score vector $z_{\text{spec}} \in \mathbb{R}^K$. Its corresponding classification probability is

$$p_{\text{spec}} = \text{Softmax}(z_{\text{spec}}) = \frac{\exp(z_{\text{spec}})}{\sum_{k=1}^K \exp(z_{\text{spec}}^{(k)})} \quad (10)$$

where K denotes the number of categories; $z_{\text{temporal}}^{(k)}$ and $z_{\text{spec}}^{(k)}$ denote the score of the k -th category; and the output vectors p_{temporal} , p_{spec} denote the predicted probability distributions of the two branches for the samples belonging to each category, respectively. Eventually, the predictions of the two branches are combined by weighted averaging to obtain the output probability vector of the overall model:

$$p_{\text{fusion}} = \lambda \cdot p_{\text{temporal}} + (1 - \lambda) \cdot p_{\text{spec}}, \quad \lambda \in [0, 1] \quad (11)$$

where λ is an adjustable fusion weight parameter to control the relative contribution of the two branches in the final decision. In this paper, λ is set to 0.5. This indicates that the two branches are weighted equally to participate in the decision. The final fusion result $p_{\text{fusion}} \in \mathbb{R}^K$ is the predicted output of the model. It is used to determine the category to which the sample belongs.

This fusion strategy has the advantages of decoupled training, efficient computation, and flexible expansion. It enables the two subnetworks to be optimized independently in different feature spaces. This prevents the problems of dimensional mismatch and information redundancy that feature-level fusion may cause. To comprehensively evaluate how the fusion method impacts model performance, this study also conducts a comparison between feature-level fusion and the hybrid fusion strategy.

3.5. Loss function

Sleep stage classification using single-lead ECG presents two critical challenges that necessitate careful loss function design. First, the dataset exhibits severe class imbalance, with minority stages (particularly N1) comprising less than 5% of total epochs, while dominant stages account for over 40%. Second, there exists significant inter-class confusion between adjacent sleep stages, requiring the model to learn discriminative features for similar patterns. Traditional cross-entropy loss tends to be dominated by majority classes and may overlook minority stages, leading to suboptimal performance on challenging but clinically important stages like N1. To address these challenges, we selected the multi-label asymmetric loss (ASLoss) function as the main training objective

function [57]. ASLoss performs effectively when confronted with category imbalance, the distribution of positive and negative samples is extremely unbalanced, and it can effectively inhibit the dominant influence of convenient-to-classify samples and simultaneously enhance the model's attention and learning capability for a few class labels. The core concept of ASLoss is to adjust the weights of the loss function by introducing asymmetric focusing and probability margin shifting to enhance the learning of minority class samples. Specifically, ASLoss uses different focusing factors γ_+ and γ_- to control the weights of positive and negative samples. It also introduces an adjustable boundary parameter p_m to crop the high-confidence negative samples so as to reduce their impact on the loss.

$$\mathcal{L}(p, y) = -[\mathbf{w}_+ \cdot \log(p) + \mathbf{w}_- \cdot \log(1 - p_m)] \quad (12)$$

where the predicted probability of the model for the positive class is denoted as p , and the probability of the negative class corrected by probabilistic boundary translation is defined as $p_m = \max(p - m, 0)$. Here, $m \in [0, 1]$ is an adjustable hyperparameter. It was set to $m = 0.05$ in this study. The weight of the positive class samples is denoted as $w_+ = (1 - p)^{\gamma_+}$, and that of the negative class samples is $w_- = (p_m)^{\gamma_-}$. These are controlled to focus on the samples of the positive and negative classes by the focusing factors γ_+ and γ_- , respectively. To enhance the model's focus on the positive class and reduce the interference of the high-confidence samples of the negative class, this study sets $\gamma_+ = 1$, $\gamma_- = 4$.

To verify the effectiveness of ASLoss, we conducted comparative experiments with additive weighted cross-entropy loss (AWCE Loss), Focal Loss, and PolyLoss. Among these, Focal Loss [58] dynamically reduces the loss weight of easy-to-classify samples by introducing a modulation factor to alleviate the adverse impact of category imbalance in model training. PolyLoss introduces a polynomial perturbation term on top of the standard cross-entropy loss to optimize the representation and generalization performance of the loss function [59].

4. Experiments

In this section, we first introduce the experimental datasets and relevant data preprocessing procedures. Next, we detail the implementation specifics, including the backbone architectures, hardware specifications, and evaluation metrics. Finally, we present the experimental results. We validated the superiority and stability of the proposed method through comprehensive analysis and comparisons.

4.1. Dataset source

Both sleep stage classification dataset and sleep apnea dataset were used in this study. This section first provides a detailed description of these datasets and then presents the specific data processing methods. The datasets used are as follows:

- 1) **SHHS** [60]: The dataset collected long-term sleep and cardiovascular health data from 6441 participants across two main data collection phases. The participants underwent an all-night PSG recording in a sleep laboratory and provided detailed clinical and lifestyle information. The experiment used SHHS Visit1 (SHHS1). Herein, each recording contained a continuous bipolar lead ECG with a sampling rate of 125 Hz. Because the subjects suffered from cardiovascular and coronary artery disease, to minimize these effects, we selected 329 subjects considered to be regular sleepers from a total of 6441 subjects in SHHS1. In addition, we included W (30 minutes before and after sleep) in the categorization of sleep stages. Stages S3 and S4 were combined as N3. Unknown periods were excluded.
- 2) **ISRUC-S3** [23]: This dataset is the third subset of the ISRUC database. It was obtained by the Sleep Medicine Center at the University Hospital of Coimbra. It includes data from 10 healthy subjects, (9 males and 1 female) with ages ranging from 30 to 58 years.

Table 1
Sleep stage and sleep apnea distributions in various datasets.

Datasets	Channels	Sampling Rate	Sleep Stage					Sleep Apnea		Total Sample
			W	N1	N2	N3	REM	Normal	Apnea	
SHHS1	ECG	125 Hz	46107	10,296	142,055	60,148	65,920	–	–	325426
ISRUC-S3	ECG	200 Hz	14.2 %	3.2 %	43.8 %	18.5 %	20.3 %	–	–	100 %
			1651	1215	2609	2014	1060	–	–	8549
			19.3 %	14.2 %	30.5 %	23.6 %	12.4 %	–	–	100 %
Apnea-ECG	ECG	100 Hz	–	–	–	–	–	20,691	12,963	33654
–	–	–	–	–	–	–	–	61.5 %	38.5 %	100 %

The PSG recordings feature 6 EEG channels, 3 EMG channels, 2 EOG channels, and 1 ECG channel. All signals in this dataset were sampled at 200 Hz and categorized into five sleep stages based on the AASM standard.

- 3) **Apnea-ECG [24]**: The Apnea-ECG database is used for sleep apnea research. It contains multiple sleep recordings from diverse populations [61]. Available on PhysioNet, it was used to validate the adaptability of the proposed algorithm across various tasks. The database consists of 70 records including a training set of 35 records (a01-a20, b01-b05, and c01-c10) and test set of 35 records (01-35). The duration of these records ranges from negligibly less than 7 h to nearly 10 h. Each record in this study contained a continuous single-lead ECG signal sampled at 100 Hz.

4.2. Dataset preprocessing

Following the guidelines set by the AASM, the physiological signals within the PSG dataset were segmented into 30 s windows. The single-lead ECG signal to be utilized was processed as follows:

- The overnight sleep data for each subject were segmented into 30 seconds time series. Specifically, we focused on selecting the data from the ECG channels and assigned labels to each sleep period based on the corresponding time intervals.
- First, the data from the selected channels were meticulously aligned with the corresponding sleep label set to remove sections that were not effectively synchronized. Second, the sleep data for each night were divided into segments of approximately 30 seconds each. If the data were not evenly divisible, the remaining portion that could not be split equally was discarded. Finally, any data without valid sleep label but incorrectly labeled as having a sleep label were eliminated and classified as unknown.
- Given that this study focuses on sleep processes, data from non-sleep periods are less meaningful. To maximize the selection of data from the sleep period, the acquisition time was intercepted: from 30 min after turning off the lights to the end of the acquisition 30 min before turning on the lights.
- To ensure signal quality and reduce the impact of noise and artifacts, the ECG data were subjected to basic quality control procedures. Specifically, we removed segments that exhibited flatline behavior, extreme amplitude fluctuations beyond physiological ranges, or implausible heart rate dynamics. In addition, segments containing large portions of missing data or obvious motion-related interference were excluded.

The data were preprocessed according to the above four points. The ECG signals in the experiments remained in their original form without formal modifications. The distribution of each stage is presented in Table 1.

4.3. Implementation

In our study, we conducted our experiments on four Nvidia RTX 4090 GPUs. The network was implemented using the PyTorch framework, with Python version 3.9.13 and Torchvision 0.21.0. The model

was trained on the SHHS1, ISRUC-S3, and Apnea-ECG datasets. To comprehensively evaluate the performance and generalizability of Sleep-ECGFusion, we employed a dual-level validation strategy. First, we conducted five-fold cross-validation on the SHHS1 and ISRUC-S3 datasets to assess the model's learning capability and stability within each dataset. This approach ensures maximum utilization of available data while providing reliable performance estimates under consistent data distributions. Second, to rigorously evaluate cross-population generalizability, we performed subject-independent cross-dataset benchmarking, training on the entire SHHS1 dataset and testing on the entire ISRUC-S3 dataset with no subject overlap. During training, a batch size of 32 and 100 epochs were used. The learning rate for the Adam optimizer was set to 0.00001. To mitigate the risk of overfitting, we applied L2 regularization. Additionally, cross-entropy loss was used as the baseline loss function for model design in the experiments.

4.4. Evaluation index

The evaluation indicators of the model are diverse. Therefore, we used the most representative metrics to describe the performance of this framework. These included the overall accuracy (ACC), macro-averaged F1 score (MF1), and Cohen kappa coefficient (κ). The relevant expression is as follows:

$$ACC = \frac{TP + TN}{TN + FP + TP + FN} \quad (13)$$

where true positive (TP) and true negative (TN) represent the number of samples predicted correctly by the model, and false positive (FP) and false negative (FN) denote the number of samples predicted incorrectly by it.

$$MF1 = \frac{1}{N} \sum_{i=1}^N F_{1i} \quad (14)$$

where N indicates the number of categories, and i suggests a certain type.

$$\kappa = \frac{ACC - P_e}{1 - P_e} \quad (15)$$

where P_e represents the accuracy of the classifier in the case of random classification. It is an indication of the random variance between the forecast and actual results of the classifier.

5. Result

5.1. Overall results

In this study, the proposed SleepECGFusion model was evaluated thoroughly in two main tasks: sleep stage classification task on the SHHS1 and ISRUC-S3 datasets, and sleep apnea detection task by transfer learning on the Apnea-ECG dataset. Table 2 summarizes the performance metrics of the model on each dataset.

Table 2

Overall Performance of SleepECGFusion across Different Evaluation Strategies. The table presents sleep staging results using five-fold cross-validation, cross-dataset benchmarking (SHHS1→ISRUC-S3), and transfer learning performance for sleep apnea detection.

Datasets	Subjects	Methods	F1-Score(%)							ACC (%)	MF1 (%)	κ
			W	N1	N2	N3	REM	Normal	Apnea			
SHHS1	111	5 -Fold	85.0	34.5	80.9	76.2	83.1	–	–	79.8	71.9	0.71
ISRUC-S3	10	5 -Fold	89.5	41.5	75.4	85.8	72.4	–	–	78.1	72.9	0.70
		Cross -Dataset	86.3	35.0	76.6	83.2	71.3	–	–	77.4	70.5	0.68
Apnea-ECG	70	Transfer Learning	–	–	–	–	–	85.0	80.0	88.4	84.8	0.70

First, in the sleep stage classification task, the model demonstrated a robust performance on the SHHS1 dataset. It performed well in terms of the F1-Score in the major sleep stages: 85.0 % in stage W, 80.9 % in stage N2, 76.2 % in stage N3, and 83.1 % in stage REM. However, the F1-Score for stage N1 was relatively low at 34.5 %. This may be related to the small sample size, transitional characteristic of sleep features, and the limited resolution of ECG signals in this stage. Overall, the model had ACC of 79.8 % on SHHS1, MF1 of 71.9 %, and κ of 0.71. On the ISRUC-S3 dataset, the model maintained a high F1-Score in stage W (89.5 %) and stage N3 (85.8 %). This was notwithstanding that the smaller sample sizes maintained a high F1-Score: 75.4 % in stage N2 and 72.4 % in stage REM. Stage N1 had a marginally higher F1-Score than SHHS1 at 41.5 %. However, it was still a relatively weak link. The overall ACC of this dataset was 78.1 %, the MF1-Score was 72.9 %, and the κ was 0.70. To rigorously evaluate the generalizability of the model in different populations and address the critical challenge of subject variability, we performed a subject-independent cross-data set performance benchmarking experiment. Specifically, the model was trained exclusively on the SHHS1 dataset and tested on the entire ISRUC-S3 dataset, ensuring no overlap of subjects between training and testing sets. The results demonstrate robust generalization capability, achieving an overall accuracy of 77.4 %, MF1 of 70.5 %, and κ of 0.68 on the independent test set (ISRUC-S3). Despite differences in subject demographics, recording devices, and signal characteristics across datasets, the model maintained performance comparable to its in-dataset evaluation. This cross-dataset validation confirms the model's robust generalization capability.

Second, to validate the cross-task generalization capability of the model, we performed transfer learning on SleepECGFusion on the Apnea-ECG dataset. This dataset focuses on sleep apnea detection and classifies sleep states into two categories: normal and apnea. After pre-training based on sleep stage classification, the model achieved effective results on the Apnea-ECG dataset: the F1-Score attained 85.0 % for the normal stage and 80.0 % for the apnea stage. The overall ACC improved significantly to 88.4 %, the MF1-Score was 84.8 %, and κ was 0.70. This result fully demonstrates that the SleepECGFusion framework displays strong knowledge migration capabilities; can migrate effective features learned from the sleep stage classification task to the sleep apnea detection task; and displays remarkable classification performance in clinically relevant tasks, particularly in scenarios with relatively limited sample size or incomplete labeling.

Overall, SleepECGFusion performs robustly on standard sleep stage classification tasks and successfully expands its application in the field of sleep apnea detection through transfer learning, verifying its effectiveness and robustness across different sleep-related tasks. However, as shown in Table 2, N1 stage exhibits notably lower performance across all epoch lengths compared to other stages. This reflects N1's inherent challenges: it represents only 3.2 % of training epochs creating severe class imbalance, exhibits intermediate ECG features between Wake and N2 making boundaries ambiguous, and shows high inter-individual variability. Despite this limitation, our method achieves strong performance on clinically critical stages.

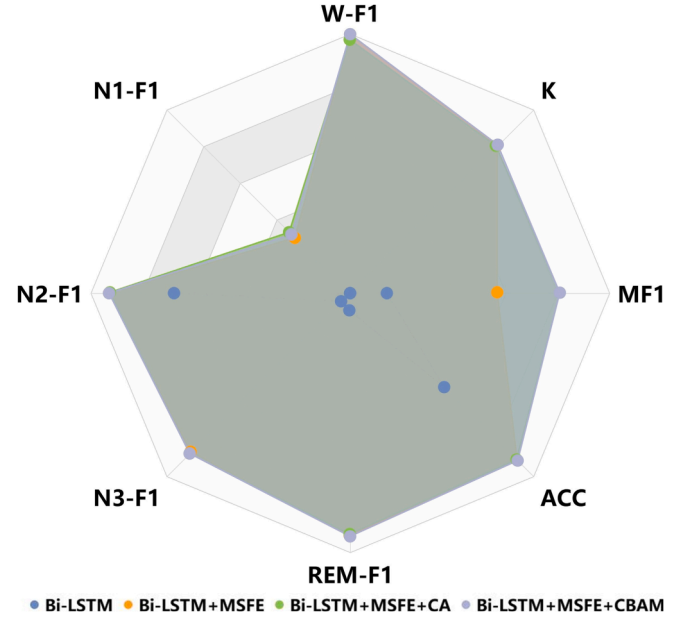


Fig. 5. Radar chart displaying ablation study results for sequence modules. Compares performance across multiple metrics under different model configurations.

5.2. Ablation experiments

To assess the contribution of each module to the performance of the model, we performed multiple sets of ablation experiments for SleepECGFusion. First, ablation was performed for the sequence part. Additionally, the role of Bi-LSTM, MSFE module, and CBAM module was analyzed. Second, the effect of image features on the model was evaluated using EfficientNet and different time-frequency plots. Finally, we conducted ablation experiments using different loss functions combined with sequence and image parts to evaluate their effects on the model performance. Through these ablation experiments, we validated the effectiveness of the modules and design options and provided data support for model optimization.

Sequence branch: A comprehensive ablation study was conducted on the SleepECGFusion sequence branch to meticulously evaluate the individual contribution of each component to the overall performance of the model. The results are presented in Table 3, where $\times$ denotes the absence and $\checkmark$ denotes the presence of a component. Initially, when only the Bi-LSTM module was present, the model exhibited an exceptionally low performance, with F1-Scores near 0 for the W and N1 stages, 57.8 % for N2, and significantly low for N3 (4.1 %) and REM (6.0 %). This resulted in a low overall ACC of 43.7 % and κ of 0. This highlights the severe limitation of relying only on Bi-LSTM for raw ECG sequence feature extraction without effective initial feature processing.

Table 3

Results of ablation experiments on sequence branch. (X: denote the absence ✓: denote the presence of a component. The best performance is highlighted in bold and underlined. Bold highlighting comes next, followed by underlining).

Bi-LSTM	MSFE	CBAM		F1-Score(%)					ACC (%)	MF1 (%)	κ
		Channel Attention	Spatial Attention	W	N1	N2	N3	REM			
✓	X	X	X	0	0	57.8	4.1	6.0	43.7	12.2	0
✓	✓	X	X	84.8	25.7	79.1	73.8	79.7	77.7	48.6	0.67
✓	✓	✓	X	84.3	28.2	78.6	74.4	79.2	77.4	68.9	0.68
✓	✓	✓	✓	84.8	27.1	79.4	74.4	79.8	77.9	69.1	0.69

Table 4

Results of ablation experiments on TF diagram map branch. (X: denote the absence ✓: denote the presence of a component. The best performance is highlighted in bold and underlined. Bold highlighting comes next, followed by underlining).

EfficientNet-B0	Methods				F1-Score(%)					ACC (%)	MF1 (%)	κ
	GASF	GADF	HHT	CWT	W	N1	N2	N3	REM			
✓	✓	X	X	X	55.8	1.9	62.1	40.6	40.4	52.7	40.1	0.30
✓	X	✓	X	X	61.1	3.3	63.8	46.9	46.1	56.1	44.2	0.36
✓	X	X	✓	X	75.3	15.2	73.0	66.1	65.9	69.5	59.1	0.56
✓	X	X	X	✓	82.0	27.8	78.6	74.3	79.2	77.1	68.4	0.68

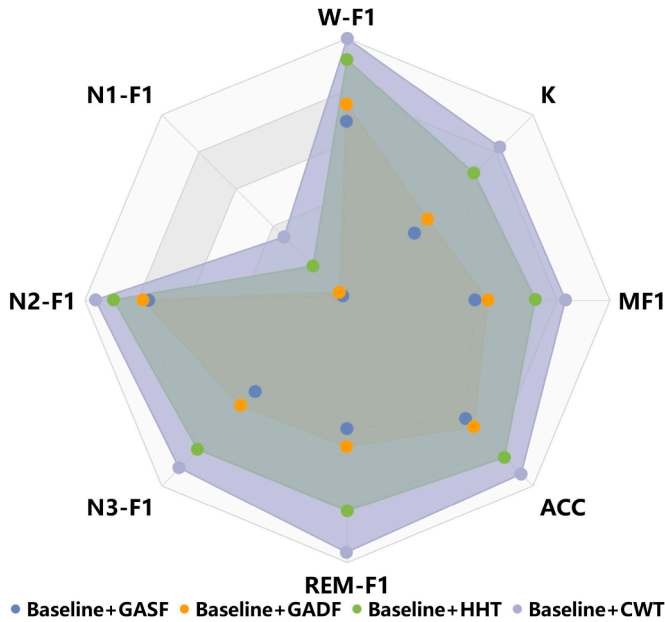


Fig. 6. Radar chart displaying ablation study results for TF diagram branch. Compares performance across multiple metrics under different model configurations.

With regard to the MSFE module, a dramatic improvement in performance was observed across all metrics. The F1-Scores for the W, N2, N3, and REM stages increased significantly to 84.8%, 79.1%, 73.8%, and 79.7% respectively. Meanwhile, the N1 F1-Score (although still challenging) improved from 0% to 25.7%. This substantial enhancement, alongside significant increases in overall ACC (to 77.7%), MF1-Score (to 68.4%), and κ (to 0.67), emphasizes the critical role of MSFE in capturing robust multi-scale features from the raw ECG signal.

Furthermore, the integration of the channel attention module within the CBAM framework resulted in a subtle albeit significant refinement to the model's performance. The N1 F1-Score increased slightly from 25.7% to 28.2%. The overall MF1-Score showed a modest increase from 68.4% to 68.9% (with κ at 0.68). This indicates that channel attention contributes by selectively highlighting more salient feature channels. Thereby, it improves the model's focus on discriminative information relevant to sleep stages.

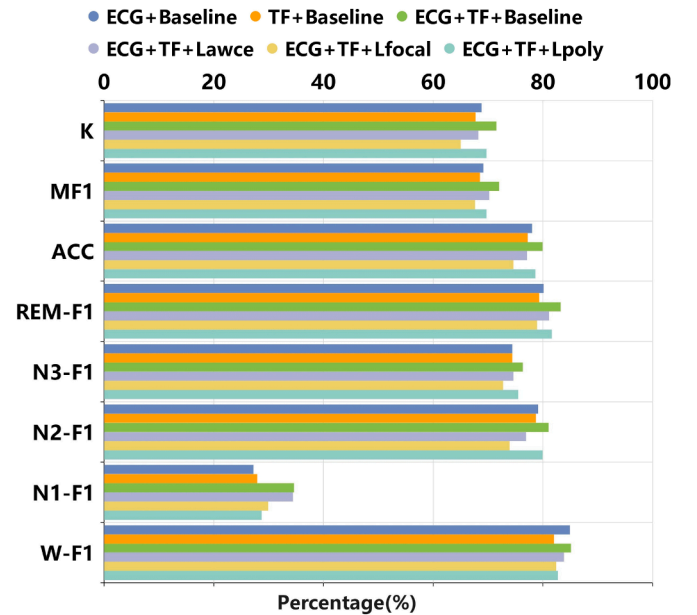


Fig. 7. Plots of the results of the overall ablation experiments of SleepECGFusion, with statistics of each performance index for different branch combinations at different loss functions.

Finally, by incorporating the full CBAM module (including both channel and spatial attention), the model achieved its maximum overall performance metrics: ACC of 77.9%, MF1-Score of 69.1%, and κ of 0.69. This complete configuration excelled overall, and the N2 F1-Score improved marginally to 79.4%. However, it is noteworthy that the N1 F1-Score decreased marginally to 27.1% compared with the channel attention-only configuration (28.2%), as shown in Fig. 5. This indicates that although spatial attention provides marginal gains in overall discrimination by focusing on spatial regions, its specific impact on the challenging N1 stage can vary. Collectively, these ablation results demonstrate that the MSFE module is essential for effective multi-scale feature extraction. It provides the most substantial performance gain. The CBAM (through its channel and spatial attention mechanisms), further refines the feature representations. Thereby, it contributes to the model's robustness and overall classification accuracy across all sleep stages.

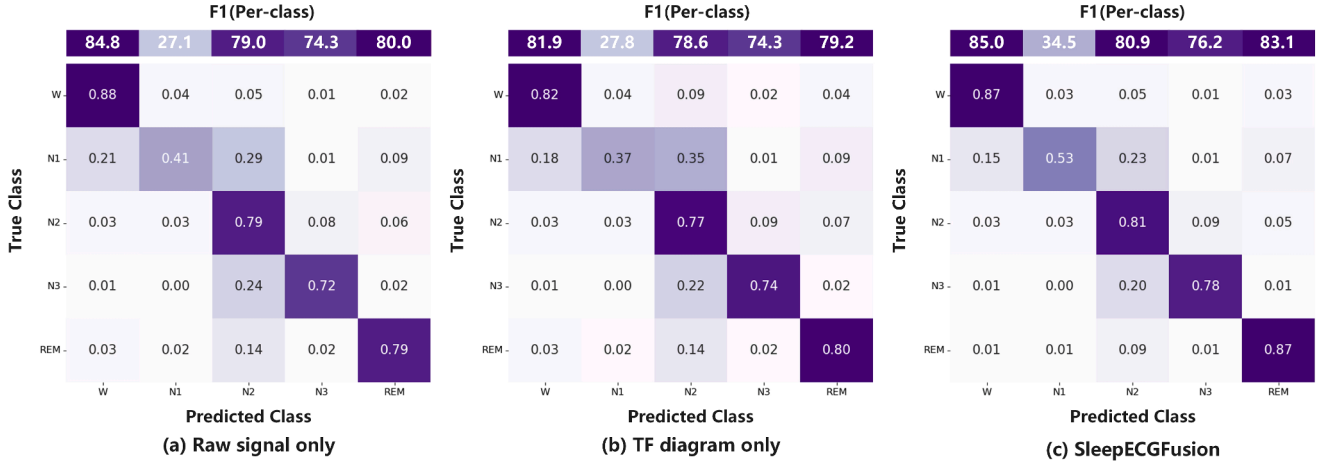


Fig. 8. Class-by-class categorized performance confusion matrix for the SleepECGFusion ablation experiment, showing the individual and joint contributions of the different modalities. (a) represents performance when using only the raw signal input, (b) represents performance when using only the time-frequency map input, and (c) represents performance when using the full SleepECGFusion framework, with the F1 scores for the corresponding sleep stages shown above each confusion matrix.

Time-frequency diagram branch: To determine the optimal time-frequency transformation method for our TF branch, an ablation study was conducted using EfficientNet-B0 as a constant backbone across four methods: GASF, GADF, HHT, and CWT. Table 4 summarizes the comparative performance across sleep stage classification metrics, where $\times$ denotes the absence and $\checkmark$ denotes the presence of a component, respectively.

The results reveal a clear progression in performance with increasing sophistication of the time-frequency representation (Fig. 6). Specifically, the model performed ineffectively with GASF (N1 F1-Score of 1.9%, ACC of 52.7%) and only marginally improved with GADF (N1 F1-Score of 3.3%, ACC of 56.1%). This indicates their limited capability to capture discriminative sleep features. A significant performance increase was observed with HHT, which significantly enhanced the N1 F1-Score to 15.2% and overall ACC to 69.5%. Crucially, the CWT consistently yielded the highest performance across all the sleep stages. It achieved an F1-Scores of 82.0% for W, 27.8% for N1, 78.6% for N2, 74.3% for N3, and 79.2% for REM; an overall ACC of 77.1%; an MF1-Score of 68.4%; and κ of 0.68. The comprehensive analysis unequivocally emphasizes CWT's superior capability in extracting robust and discriminative time-frequency features from ECG signals. This establishes it as the optimal transformation method for SleepECGFusion's TF branch.

SleepECGFusion: A comprehensive set of ablation experiments was conducted to assess the combined impact of the components of the proposed SleepECGFusion architecture and the selected loss functions on the sleep stage classification performance. Table 5 details these results. The "Baseline" specifically refers to ASLoss developed in this study. $\times$ and $\checkmark$ denote the absence and presence of a component, respectively. Initially, models utilizing only one modality (ECG input or TF input) showed moderate performances. For example, the ECG-only model achieved an overall ACC of 77.9% and MF1-Score of 69.1%, whereas the TF-only model attained ACC of 77.1% and an MF1-Score of 68.4%. Significantly, the N1 stage remained challenging for both the single-modality inputs.

Crucially, the full SleepECGFusion framework (incorporating both ECG and TF inputs with ASLoss) demonstrated superior performance. This configuration achieved the highest overall ACC of 79.8%, an MF1-Score of 71.9%, and a κ of 0.71. It also yielded strong F1-Scores across all the individual sleep stages, including 85.0% in W, 34.5% in N1, 80.9% in N2, 76.2% in N3, and 83.1% in REM. This significant improvement unequivocally validated the synergistic effect of integrating both temporal and time-frequency representations.

To systematically validate our choice of ASLoss, we conducted comparative experiments with three representative loss functions, each addressing different aspects of the imbalanced classification problem: $\mathcal{L}_{AWCE}$ serves as the baseline, using class-frequency-based weighting to handle imbalance. However, it treats all samples within each class equally, without differentiating between easy and hard examples. $\mathcal{L}_{Focal}$ addresses imbalance through dynamic sample reweighting based on prediction confidence. While effective in object detection, it applies symmetric focusing to both positive and negative classes, which may not be optimal for extremely imbalanced sleep data. $\mathcal{L}_{Poly}$ introduces polynomial perturbation on cross-entropy to optimize the loss landscape. It provides a complementary perspective by modifying optimization dynamics rather than sample weighting. A comparative analysis of the loss functions within the dual-modality setup revealed that ASLoss consistently outperformed $\mathcal{L}_{AWCE}$, $\mathcal{L}_{Focal}$, and $\mathcal{L}_{Poly}$ in terms of overall metrics. In contrast, $\mathcal{L}_{Focal}$ and $\mathcal{L}_{Poly}$ generally resulted in decreased performance across most stages. A significant difference can be observed in Fig. 7. The classification performance of the confusion matrices for different branches is displayed in Fig. 8. From the results of the confusion matrix, it is evident that SleepECGFusion obtained the best classification performance. These comprehensive comparisons robustly verify that ASLoss provides the best overall performance within the SleepECGFusion framework, as it optimally balances the accuracy across all sleep stages and ensures robust model learning for imbalanced sleep data. These results collectively affirm that the proposed SleepECGFusion architecture, particularly when coupled with ASLoss, achieves optimal balance across sleep stages and overall classification robustness. This makes it the most effective configuration.

6. Discussion and analysis

In this section, we focus on the following aspects: First, we compare the strengths and weaknesses of our proposed framework with those of existing methods, particularly in terms of the accuracy of sleep stage classification. Second, we evaluate the effects of different fusion methods on the model performance and evaluate its performance using SHHS1 as a test dataset. Third, we analyze the effects of different duration inputs on the performance of the model, particularly the difference between the short and long time window contribution to sleep stage classification. Then, we discuss the performance differences of the framework in sleep apnea populations versus healthy populations and make recommendations for further improvements. Finally, we conduct further investigations into the model's interpretability.

Table 5

Results of ablation experiments with SleepECGFusion. (X: denote the absence ✓: denote the presence of a component. The best performance is highlighted in bold and underlined. Bold highlighting comes next, followed by underlining).

Input Type	Loss				F1-Score (%)					ACC (%)	MF1 (%)	κ
	Baseline	$\mathcal{L}_{AWCE}$	$\mathcal{L}_{Focal}$	$\mathcal{L}_{Poly}$	W	N1	N2	N3	REM			
ECG input	✓	X	X	X	84.8	27.1	79.0	74.3	80.0	<u>77.9</u>	<u>69.1</u>	<u>0.69</u>
TF input	✓	X	X	X	81.9	27.8	<u>78.6</u>	74.3	79.2	77.1	68.4	0.68
ECG input + TF input	✓	X	X	X	85.0	34.5	80.9	76.2	83.1	79.8	71.9	0.71
ECG input + TF input	X	✓	X	X	<u>83.7</u>	34.3	76.8	<u>74.5</u>	<u>81.0</u>	77.0	70.1	0.68
ECG input + TF input	X	X	✓	X	82.3	29.8	73.8	72.6	78.8	74.5	67.5	0.65
ECG input + TF input	X	X	X	✓	82.6	<u>28.6</u>	79.8	75.4	81.5	78.5	69.6	0.70

Table 6

Performance comparison with the cutting-edge methods on these three datasets.

Dataset	Methods	Sensing Modality	Signals/Features	Sleep Stages	Acc(%)	κ	MF1(%)
MITBPD	SelfAttentionNet (2024) [62]	ECG	ECG	5	<u>80.6</u>	0.68	70.5
	SELF DCRNet (2022) [63]	ECG	HRV	5	74.2	-	61.6
	MESA Resnet (2023) [64]	ECG	ECG, IHR	5	77.3	-	-
	MESA LSTM (2025) [46]	ECG	ACT, HRV, RRV	5	65.7	-	60.1
	ISRUC-S3 LDANet (2023) [65]	ECG,EEG	ECG,EEG	5	76.1	0.68	72.9
	ISRUC-S3 **This Study (SleepECGFusion)	ECG	ECG	5	78.1	0.70	72.9
	SHHS1 CNN + LSTM (2020) [47]	ECG	ECG, Respiratory	5	72.5	-	-
	SHHS1 DeepAdaptationNet (2022) [50]	ECG	ECG	4	74.8	0.67	-
	SHHS1 DualStream Model (2023) [66]	ECG	ECG	4	73.3	0.62	<u>73.0</u>
	SHHS1 **This Study (SleepECGFusion)	ECG	ECG - 30S	5	79.8	<u>0.71</u>	71.9
Apnea-ECG	CNN + RNN (2021) [67]	ECG	ECG	-	80.6	-	<u>74.7</u>
	CNN + DRNN (2022) [68]	ECG	ECG	-	<u>88.1</u>	-	84.0
	Hybrid Transformer (2022) [69]	ECG	ECG	-	72.3	-	68.9
	CNN + Transformer (2023) [70]	ECG	ECG	-	88.2	-	-
	OSAformer (2025) [71]	ECG	ECG	-	82.4	-	-
	**This Study (SleepECGFusion)	ECG	ECG	-	88.4	<u>0.70</u>	84.8
					<u>90.6</u>	<u>0.87</u>	84.2

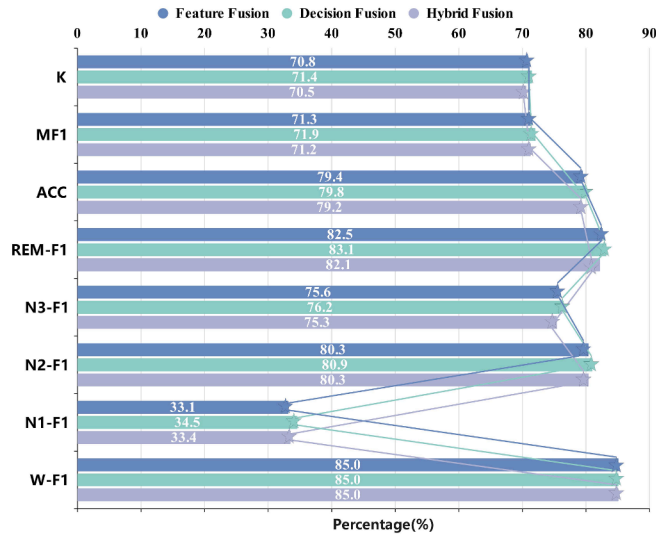


Fig. 9. Comparison of performance metrics under different fusion methods, where stacked bar charts count the values of each metric and line graphs connected by different colored $\star$ visualize the changes in each metric.

6.1. Comparison with state-of-the-art methods

To ascertain the competitive performance of SleepECGFusion, we conducted a comprehensive comparison against cutting-edge methods

across three benchmark datasets. Table 6 summarizes these comparative results. It highlights the superiority of our method.

On the ISRUC-S3 dataset, SleepECGFusion had ACC of 78.1 %, κ of 0.70, and MF1 of 72.9 %. This performance is an improvement over those of existing single-lead ECG-based methods such as LDANet (ACC 76.1 %, κ 0.68, and MF1 72.9 %) This highlights the strong foundation of our framework for standard sleep stage classification. For the SHHS1 dataset, SleepECGFusion demonstrated remarkable advancements, particularly with optimized input durations. Although our model with 30 s input achieved a competitive ACC of 79.8 %, κ of 0.71, and MF1 of 71.9 %, its performance increased significantly with longer segments. With 60 s inputs, SleepECGFusion attained a remarkable ACC of 86.5 %, κ of 0.81, and MF1 of 77.8 %. When extended to 90 s inputs, our method achieved a new state-of-the-art performance with a remarkable ACC of 90.6 %, κ of 0.87, and MF1 of 84.2 %. That is, conclusively outperformed all the contemporary ECG-based methods listed in SHHS1, many of which utilize shorter segments or different feature engineering approaches. This emphasizes the advantages of our cross-modal fusion and optimized input duration.

In the Apnea-ECG dataset for sleep apnea detection, SleepECGFusion (using transfer learning) also established a new benchmark. It achieved ACC of 88.4 %, κ of 0.70, and MF1 of 84.8 %. This performance is highly competitive. It surpasses those of several recent deep learning models for apnea detection on this dataset, including CNN + DRNN (ACC 88.1 %, MF1 84.0 %) and CNN + Transformer (ACC 88.2 %). This reinforces SleepECGFusion's strong generalization capabilities.

Collectively, these comparative results verify that SleepECGFusion sets a new state-of-the-art in single-lead ECG-based sleep stage classification and exhibits a robust and relatively high performance in sleep

apnea detection. This validates its effectiveness and versatility across sleep-related tasks and datasets.

6.2. Comparison of integration modalities

In this subsection, we evaluate feature-level fusion, decision-level fusion, and hybrid fusion separately. The results are shown in Fig. 9. The feature-level fusion method performs stably in the W, N2, N3, and REM stages, with the F1-Score remaining high. However, in stage N1, the F1-Score of the model is lower at 33.1 % owing to the small sample size in this stage. The insufficiency of sample size may result in the model's incapability to fully learn the features in this stage. This would affect the classification performance. In contrast, decision-level fusion effectively improves the performance of the N1 stage by integrating the prediction results of multiple models, and the F1-Score improves to 34.5 %. This indicates that decision-level fusion can mitigate the adverse impact of sample imbalance on the model performance by integrating the assessments of different models. In addition, the performance of decision-level fusion in other stages is comparable to that of feature-level fusion, with an overall ACC of 79.8 %, MF1 of 71.9 %, and κ of 0.71. The hybrid fusion method combines the advantages of feature-level fusion and decision-level fusion. Additionally, it maintains a more balanced performance in all the stages. In the N1 stage, the F1-Score of hybrid fusion is 33.4 %, which is marginally higher than that of decision-level fusion. This demonstrates the advantages of the hybrid method in addressing unbalanced data. The overall ACC is 79.2 %, MF1 is 71.2 %, and κ is 0.71. This demonstrates the capability of hybrid fusion in balancing the performance and improving the performance of specific stages. To summarize, decision-level fusion and hybrid fusion have significant advantages at stage N1. These can mitigate the effects of smaller sample sizes by integrating the predictions of different models. Feature-level fusion performs stably in most stages. However, it has certain limitations in the N1 stage where the sample size is small. Depending on the task requirements, decision-level fusion and hybrid fusion may be more suitable for addressing sample imbalance, whereas hybrid fusion provides a better overall balance.

6.3. Effect of different lengths of inputs

To evaluate the effect of different input durations on the model performance. We conducted experiments using ECG segments of different durations (Fig. 10(a)–(c)). The results show that the model's performance in multiple sleep stages improved significantly with the increase in input durations. For an input duration of 30 s, the model had F1-Scores of 85.1 % in stage W, 32.6 % in stage N1, 80.7 % in stage N2, 76.0 % in stage N3, and 82.7 % in stage REM; an overall ACC of 79.6 %; MF1 of 71.4 %; and κ of 0.71. Although the model performed better in stages W, N2, and REM, the F1-Score in stage N1 was lower at 32.6 %. This may be because the 30 s time window was too short and failed to capture sufficient information regarding the sleep cycle, thereby resulting in a lower classification accuracy in this stage. The overall performance of the model improved further when the input duration was increased to 60 s. The F1-Score increased to 90.4 % for stage W and 37.9 % for stage N1, compared with 87.1 % for stage N2, 85.5 % for stage N3, and 87.6 % for stage REM. At this point, ACC was 86.7 %, MF1 was 77.7 %, and κ was 0.81. This indicates that the model's capability for capturing the variations in sleep state increases as the input duration does, particularly in stage N1. The improvement in the F1-Score indicates that the 60 s input provides the model with more contextual information. This, in turn, improves the classification performance in that stage. The model's performance in all the sleep stages improved further when a 90 s input duration was used. In particular, the F1-Score improved to 54.5 % in stage N1. This demonstrated that the 90 s input window significantly improved the model's capability to classify this stage. The F1-Score attained 94.2 % in stage W, 91.1 % in stage N2, 89.8 % in stage N3, and 91.2 % in stage REM. Moreover, the

overall ACC, MF1, and κ were 91.0 %, 84.2 %, and 0.87, respectively. This result indicates that as the input length increases, the model is capable of capturing more temporal information. This results in a significant improvement in classification accuracy in all the stages.

The consistent N1 underperformance across experiments reflects physiological rather than methodological limitations. Single-lead ECG captures only indirect autonomic changes during this brief transitional state, where heart rate variability features occupy an intermediate space between wakefulness and stable sleep. Combined with severe class imbalance and high inter-subject variability, this creates an inherent ceiling on ECG-based N1 detection. Encouragingly, our method achieves robust performance on diagnostically prioritized stages, and the improvement from 30s to 90s epochs demonstrates effective temporal modeling. Future integration of multi-modal signals and temporal contrastive learning approaches may further address these challenges.

In summary, longer input durations have a significant positive impact on the model performance. This is particularly so in the N1 stage, where the classification accuracy improvement is particularly significant. The longer duration window provides more contextual information. This enables the model to capture dynamic changes in the sleep state more accurately, particularly when recognizing shorter sleep stages. Thus, longer input durations are evidently more effective for improving the overall performance of the model. This is particularly so when addressing complex sleep states.

6.4. Comparison of results for different groups

To investigate the differences in sleep stage classification between healthy and apneic sleep apnea populations, the experiment compared the sleep stage classification performance between healthy and apneic sleep populations at different input durations. The results reveal that the model generally performed better in the healthy population than in the sleep apnea population. This was particularly so at longer input durations, where the classification accuracy of the healthy population was improved significantly. For the healthy population (Fig. 10 (d)–(f)), the model's F1-Score improved progressively across multiple sleep stages as the input duration increased. For example, at an input duration of 30 s, the F1-Score was 85.0 % for stage W, 34.5 % for stage N1, 80.9 % for stage N2, 76.0 % for stage N3, and 83.1 % for stage REM. Meanwhile, when the input duration was increased to 60 s, the F1-Score improved to 90.4 % for the W stage, 37.9 % for the N1 stage, 87.1 % for the N2 stage, 85.5 % for the N3 stage, and 87.6 % for the REM stage. A further increase to 90 s yielded the best performance of the model in all the sleep stages, with an F1-score of 93.7 % for stage W, 56.1 % for stage N1, 90.8 % for stage N2, 89.5 % for stage N3, and 90.8 % for stage REM. This enhancement indicates that the model can capture more temporal information as the input duration increases, and it displays significant improvement particularly in the short N1 stage.

However, the results for the sleep apnea population (Fig. 10 (g)–(i)) reveal that although longer input durations improve the model performance, it still underperforms for the healthy population. At 30 s input duration, the F1-Score was 82.4 % for the W stage, 34.8 % for the N1 stage, 82.5 % for the N2 stage, 68.6 % for the N3 stage, and 74.3 % for the REM stage. As the input duration increased, the F1-Score improved from 34.8 % to 55.9 % for the N1 stage, from 82.5 % to 93.5 % for the N2 stage, from 68.6 % to 86.6 % for the N3 stage, and from 74.3 % to 89.7 % for the REM stage. Notwithstanding the significant performance improvement in stages N1 and N3, the overall F1 score was still lower than that of the healthy population. This may be related to the instability of the ECG signal and the presence of a large amount of noise and disturbances such as the apnea and blood oxygen fluctuations in patients with sleep apnea, which may have affected the capability of the model to accurately delineate sleep stages.

In general, the healthy population showed higher classification accuracies with longer input durations. This was particularly so in the deep sleep stages and REM stages, where the F1 score improved significantly.

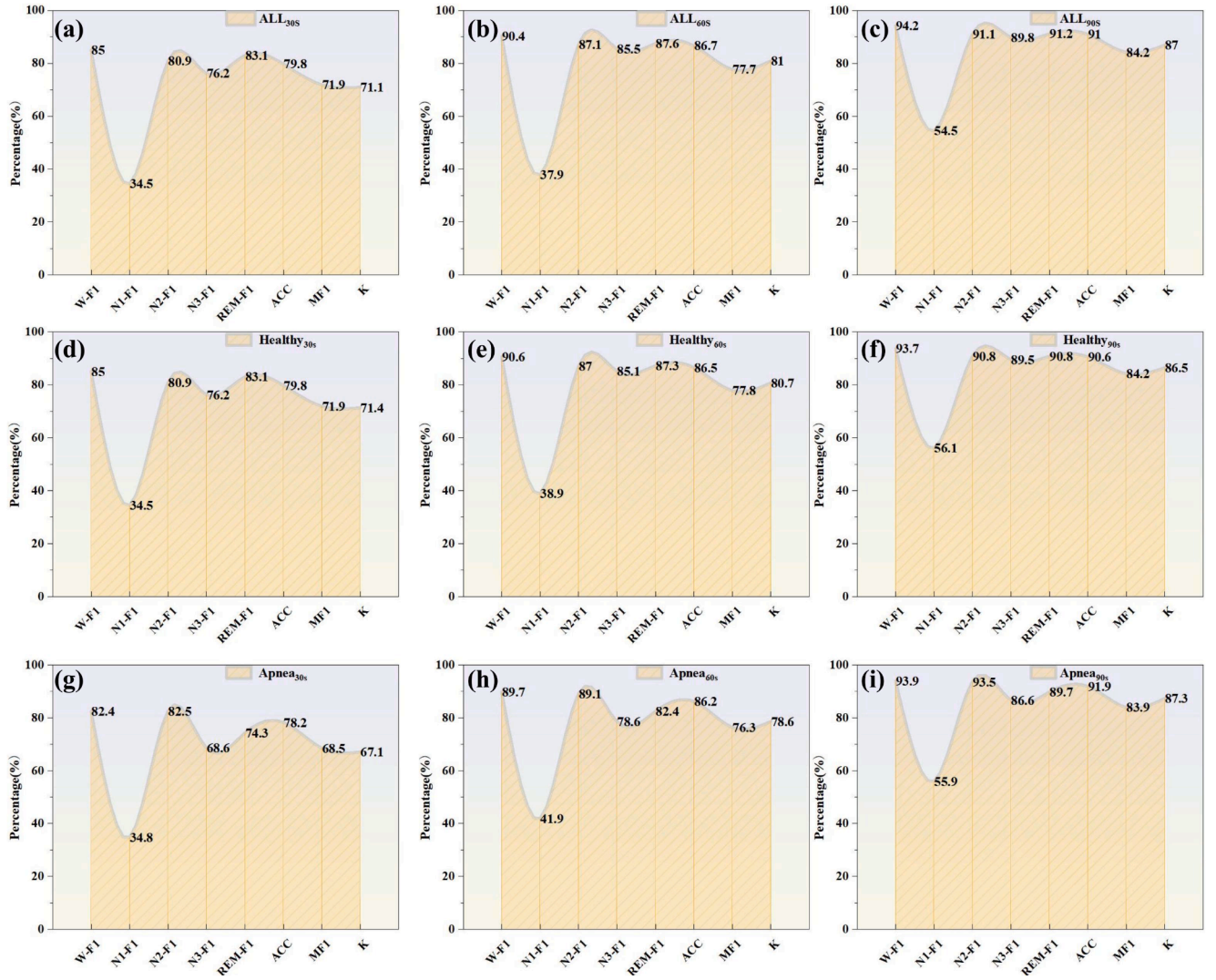


Fig. 10. Comprehensive Assessment of SleepECGFusion Models for Sleep Stage Classification Performance in Different Populations and ECG Signaling Fragment Lengths. Among them, the first rows (a-c) present the classification results of the model in all populations; the second rows (d-f) specifically evaluate the performance of the model on a subset of healthy populations; while the third rows (g-i) focus on the classification results in the sleep apnea population. Horizontally, the first columns (a, d, g) correspond to classification results using 30-second ECG signal segments; the second columns (b, e, h) are evaluated for 60-second segments; and the third columns (c, f, i) demonstrate performance under 90-second segments. Each subplot meticulously enumerates the F1 scores for each sleep stage, as well as the ACC, MFI, and K, thus visually revealing the impact of signal segment length and specific population characteristics on the model's sleep stage classification accuracy.

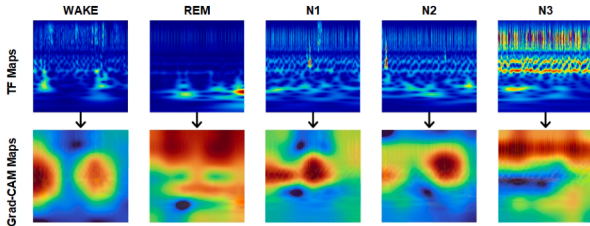


Fig. 11. Grad-CAM visualizations for different sleep stages.

For the sleep apnea population, however, although the longer input duration helped improve the model's classification accuracy in the N1, N3, and REM stages, the model performance was still lower than that of the healthy population owing to the frequent apnea events during sleep.

6.5. Model interpretability analysis

To enhance the interpretability of the proposed framework and facilitate its potential clinical application, we incorporated explainability techniques that help reveal the decision-making process of the model. Specifically, two complementary methods were employed: Gradient-weighted Class Activation Mapping (Grad-CAM) and Uniform Manifold Approximation and Projection (UMAP). The Grad-CAM visualization results indicate that the model's attention patterns show basic consistency with known physiological mechanisms across different sleep stages: focusing on low-frequency features during Wake, covering broader mid-to-high frequency components in REM, showing dispersed attention in N1, concentrating on mid-low frequencies in N2, and emphasizing low-frequency signals in N3, which corresponds to parasympathetic dominance and reduced, stabilized heart rate during deep sleep. These findings suggest that the model captures physiologically relevant features, providing a certain degree of interpretability, as shown in Fig. 11.

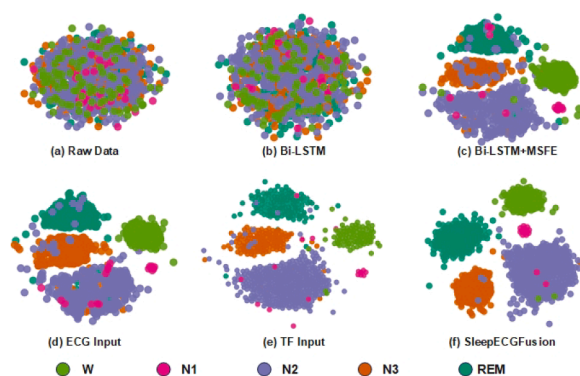


Fig. 12. UMAP visualizations for different sleep stages.

To qualitatively evaluate the separability of sleep stages in the learned feature space, we applied UMAP to visualize the high-dimensional representations generated by different model variants. As shown in Fig. 12, baseline models such as raw input-based models and Bi-LSTM produce highly entangled clusters with poor class boundaries. In contrast, our proposed SleepECGFusion framework exhibits notably improved clustering structure, with sleep stages showing clearer spatial organization and less overlap. Although some boundaries remain partially ambiguous, the overall separation across the five stages is enhanced, including the N1 stage, where classification performance has notably improved compared to the baseline models. This suggests that SleepECGFusion is more effective in learning stage-specific and physiologically meaningful embeddings compared to baseline approaches.

7. Conclusion

This study presents an innovative cross-modal deep learning framework called SleepECGFusion, which successfully integrates information from two complementary domains. Furthermore, this study verifies the conjecture that increasing input signal duration consistently improves classification accuracy across all sleep stages—particularly for the frequently misclassified N1 stage. This highlights the value of a richer temporal context. SleepECGFusion achieved state-of-the-art performance in ECG-based sleep stage classification tasks, and demonstrated superior migration capabilities. Crucially, the framework provided robust and comparable classification results for both healthy individuals and sleep apnea patients. Future research will be aimed at further optimizing SleepECGFusion for clinical deployment. This will include incorporating additional physiological signals to enhance its robustness in highly perturbed sleep, exploring temporal contrastive learning approaches to better address temporal variability in long-term monitoring, and evaluating its application in other sleep-related neurological disorders.

CRedit authorship contribution statement

Xuanhao Qi: Writing - original draft, Validation, Data curation. **Li Chen:** Writing - review & editing, Formal analysis, Conceptualization. **Hongze Liu:** Formal analysis, Conceptualization. **Shishi Tang:** Errata. **Zhipeng He:** Syntax check. **Yichen Dai:** Data processing, **Kaizhe Zheng:** Investigation. **Jianping Man:** Methodology. **Yi Zhou:** Supervision.

Ethical statement

This article does not include any experimental studies involving humans or animals. All datasets used in this paper are public datasets. Therefore, our research does not need an Ethics statement.

Data availability

All data used in this study are publicly available.

CRedit authorship contribution statement

Xuanhao Qi: Writing – original draft, Methodology; **Li Chen:** Conceptualization; **Hongze Liu:** Investigation, Data curation, Conceptualization; **Shishi Tang:** Supervision, Project administration, Investigation; **Zhipeng He:** Validation, Supervision; **Yichen Dai:** Visualization; **Kaizhe Zheng:** Methodology, Conceptualization; **Jianping Man:** Supervision; **Yi Zhou:** Supervision, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] P. Huang, M. Qiu, Y. Liu, X. Fan, Advancing sleep stages classification through a dual-graphormer approach, *Expert Syst. Appl.* (2025) 128220.
- [2] Y. Lou, L. Zhou, F. Song, F. Xu, X. Wang, W. Li, Z. Qian, A novel sleep modulation and quality classification strategy based EEG signals and multi-scale attention deep learning, *IEEE Internet Things J.* (2025).
- [3] M.J. Sateia, International classification of sleep disorders-Third edition, *Chest* 146 (5) (2014) 1387–1394. <https://doi.org/https://doi.org/10.1378/chest.14-0970>
- [4] H. Lee, Y.R. Choi, H.K. Lee, J. Jeong, J. Hong, H.-W. Shin, H.-S. Kim, Explainable vision transformer for automatic visual sleep staging on multimodal PSG signals, *npj Digit. Med.* 8 (1) (2025) 55.
- [5] P. Liu, W. Qian, H. Zhang, Y. Zhu, Q. Hong, Q. Li, Y. Yao, Automatic sleep stage classification using deep learning: signals, data representation, and neural networks, *Artif. Intell. Rev.* 57 (11) (2024) 301.
- [6] O. Katz, R. Talmon, Y.-L. Lo, H.-T. Wu, Alternating diffusion maps for multimodal data fusion, *Inform. Fusion* 45 (2019) 346–360.
- [7] M. Samaee, M. Yazdi, D. Massicotte, Multi-modal signal integration for enhanced sleep stage classification: leveraging EOG and 2-channel EEG data with advanced feature extraction, *Artif. Intell. Med.* (2025) 103152.
- [8] X. Xu, F. Cong, Y. Chen, J. Chen, Sleep stage classification with multi-modal fusion and denoising diffusion model, *IEEE J. Biomed. Health Inform.* (2024).
- [9] J. Duan, J. Xiong, Y. Li, W. Ding, Deep learning based multimodal biomedical data fusion: an overview and comparative review, *Inform. Fusion* (2024) 102536.
- [10] R. Thapa, B. He, M.R. Kjaer, H. Moore, IV, G. Ganjoo, E. Mignot, J.Y. Zou, Sleepfm: multi-modal representation learning for sleep across ecg, eeg and respiratory signals, in: AAAI 2024 Spring Symposium on Clinical Foundation Models, 2024.
- [11] C. Zhao, W. Wu, H. Zhang, R. Zhang, X. Zheng, X. Kong, Sleep stage classification via multi-view based self-supervised contrastive learning of EEG, *IEEE J. Biomed. Health Inform.* (2024).
- [12] P. An, Z. Yuan, J. Zhao, X. Jiang, B. Du, An effective multi-model fusion method for EEG-based sleep stage classification, *Knowl. Base. Syst.* 219 (2021) 106890.
- [13] R.V. Sharan, H. Takeuchi, A. Kishi, Y. Yamamoto, Macro-sleep staging with ECG-derived instantaneous heart rate and respiration signals and multi-input 1D CNN-BiGRU, *IEEE Trans. Instrum. Meas.* (2024).
- [14] S.H. Mostafaei, J. Tanha, A. Sharafkhaneh, A novel deep learning model based on transformer and cross modality attention for classification of sleep stages, *J. Biomed. Inform.* 157 (2024) 104689.

- [15] K. Yang, M. Hong, J. Zhang, Y. Luo, S. Zhao, O. Zhang, X. Yu, J. Zhou, L. Yang, P. Zhang, et al., ECG-LM: understanding electrocardiogram with a large language model, *Health Data Sci.* 5 (2025) 0221.
- [16] H. Khorrami, M. Moavani, A comparative study of DWT, CWT and DCT transformations in ECG arrhythmias classification, *Expert Syst. Appl.* 37 (8) (2010) 5751–5757.
- [17] W. Ko, E. Jeon, S. Jeong, H.-I. Suk, Multi-scale neural network for EEG representation learning in BCI, *IEEE Comput. Intell. Mag.* 16 (2) (2021) 31–45.
- [18] Y. Wang, Y. Shi, Z. He, Z. Chen, Y. Zhou, Combining temporal and spatial attention for seizure prediction, *Health Inf. Sci. Syst.* 11 (1) (2023) 38.
- [19] K. Liu, X. Xing, T. Yang, Z. Yu, B. Xiao, G. Wang, W. Wu, DMSACNN: Deep multiscale attentional convolutional neural network for EEG-based motor decoding, *IEEE J. Biomed. Health Inform.* (2025).
- [20] I. Ariza, L.J. Tardón, A.M. Barbancho, I. Barbancho, EEG-based listened-language classification, *Expert Syst. Appl.* (2025) 128276.
- [21] S. Bagherzadeh, M.S. Shahabi, A. Shalbaf, Detection of schizophrenia using hybrid of deep learning and brain effective connectivity image from electroencephalogram signal, *Comput. Biol. Med.* 146 (2022) 105570.
- [22] X. Chen, Y. Chen, W. Ma, X. Fan, Y. Li, Toward sleep apnea detection with lightweight multi-scaled fusion network, *Knowl. Base. Syst.* 247 (2022) 108783.
- [23] S. Khalighi, T. Sousa, J.M. Santos, U. Nunes, ISRUC-sleep: a comprehensive public dataset for sleep researchers, *Comput. Method. Program. Biomed.* 124 (2016) 180–192.
- [24] T. Penzel, G.B. Moody, R.G. Mark, A.L. Goldberger, J.H. Peter, The apnea-ECG database, in: *Computers in Cardiology 2000*. Vol. 27 (Cat. 00CH37163), IEEE, 2000, pp. 255–258.
- [25] M. Xiao, H. Yan, J. Song, Y. Yang, X. Yang, Sleep stages classification based on heart rate variability and random forest, *Biomed. Signal Process. Control* 8 (6) (2013) 624–633.
- [26] M. Adnane, Z. Jiang, Z. Yan, Sleep–wake stages classification and sleep efficiency estimation using single-lead electrocardiogram, *Expert Syst. Appl.* 39 (1) (2012) 1401–1413.
- [27] P.K. Stein, Y. Pu, Heart rate variability, sleep and sleep disorders, *Sleep Med. Rev.* 16 (1) (2012) 47–66.
- [28] H. Phan, F. Andreotti, N. Cooray, O.Y. Chén, M. De Vos, Joint classification and prediction CNN framework for automatic sleep stage classification, *IEEE Trans. Biomed. Eng.* 66 (5) (2018) 1285–1296.
- [29] N. Sridhar, A. Shueb, P. Stephens, A. Kharbouch, D.B. Shimol, J. Burkart, A. Ghoreysy, L. Myers, Deep learning for automatic sleep staging using instantaneous heart rate, *NPJ Digit. Med.* 3 (1) (2020) 106.
- [30] H. Phan, O.Y. Chén, M.C. Tran, P. Koch, A. Mertins, M. De Vos, XSleepNet: multi-view sequential model for automatic sleep staging, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (9) (2022) 5903–5915.
- [31] R. Huttunen, T. Leppänen, B. Duce, E.S. Arnardottir, S. Nikkonen, S. Myllymaa, J. Töyräs, H. Korkalainen, A comparison of signal combinations for deep learning-based simultaneous sleep staging and respiratory event detection, *IEEE Trans. Biomed. Eng.* 70 (5) (2023) 1704–1714.
- [32] E. Fuster-García, A. Bresó, J. Martínez-Miranda, J. Rosell-Ferrer, C. Matheson, J.M. García-Gómez, Fusing actigraphy signals for outpatient monitoring, *Inform. Fusion* 23 (2015) 69–80.
- [33] J. Liu, S. Lin, B.W.-K. Ling, DMIN: Low-rank representation learning based on dynamic mode for sleep staging, *Expert Syst. Appl.* 279 (2025) 127424. <https://doi.org/https://doi.org/10.1016/j.eswa.2025.127424>
- [34] M. Radha, P. Fonseca, A. Moreau, M. Ross, A. Cerny, P. Anderer, X. Long, R.M. Aarts, A deep transfer learning approach for wearable sleep stage classification with photoplethysmography, *NPJ Digit. Med.* 4 (1) (2021) 135.
- [35] M. Gaiduk, J.J.P. Rodríguez, R. Seepold, N.M. Madrid, T. Penzel, M. Glos, J.A. Ortega, Estimation of sleep stages analyzing respiratory and movement signals, *IEEE J. Biomed. Health Inform.* 26 (2) (2021) 505–514.
- [36] M. Malik, J.T. Bigger, A.J. Camm, R.E. Kleiger, A. Malliani, A.J. Moss, P.J. Schwartz, Heart rate variability: standards of measurement, physiological interpretation, and clinical use, *Eur. Heart J.* 17 (3) (1996) 354–381.
- [37] V.K. Somers, M.E. Dyken, A.L. Mark, F.M. Abboud, Sympathetic-nerve activity during sleep in normal subjects, *N. Engl. J. Med.* 328 (5) (1993) 303–307.
- [38] A. Matsuyama, M. Jonkman, The application of wavelet and feature vectors to ECG signals, *Austral. Phys. Eng. Sci. Med.* 29 (1) (2006) 13.
- [39] C.-L. Liu, B. Xiao, C.-H. Hsieh, Multimodal fusion of spatial–temporal and frequency representations for enhanced ECG classification, *Inform. Fusion* 118 (2025) 102999.
- [40] J. Singh, R.K. Sharma, A.K. Gupta, A method of REM-NREM sleep distinction using ECG signal for unobtrusive personal monitoring, *Comput. Biol. Med.* 78 (2016) 138–143.
- [41] S. Satapathy, D. Loganathan, H.K. Kondaveeti, R. Rath, Performance analysis of machine learning algorithms on automated sleep staging feature sets, *CAAI Transact. Intell. Technol.* 6 (2) (2021) 155–174.
- [42] E. Alickovic, A. Subasi, Ensemble SVM method for automatic sleep stage classification, *IEEE Trans. Instrum. Meas.* 67 (6) (2018) 1258–1265.
- [43] X. Li, L. Cui, S. Tao, J. Chen, X. Zhang, G.-Q. Zhang, Hyclasss: a hybrid classifier for automatic sleep stage scoring, *IEEE J. Biomed. Health Inform.* 22 (2) (2017) 375–385.
- [44] D. Jiang, M.A. Yu, W. Yuanyuan, et al., Robust sleep stage classification with single-channel EEG signals using multimodal decomposition and HMM-based refinement, *Expert Syst. Appl.* 121 (2019) 188–203.
- [45] I. Choi, W. Sung, Sleep model: a sequence model for predicting the next sleep stage, in: 2023 IEEE Biomedical Circuits and Systems Conference (BioCAS), IEEE, 2023, pp. 1–5.
- [46] D. Krauss, R. Richer, A. Küderle, J. Jukic, A. German, H. Leutheuser, M. Regensburger, J. Winkler, B.M. Eskofier, Incorporating respiratory signals for machine learning-based multimodal sleep stage classification: a large-scale benchmark study with actigraphy and heart rate variability, *Sleep* (2025) zsa091.
- [47] H. Sun, W. Ganglberger, E. Panneerselvam, M.J. Leone, S.A. Quadri, B. Goparaju, R.A. Tesh, O. Akeju, R.J. Thomas, M.B. Westover, Sleep staging from electrocardiography and respiration with deep learning, *Sleep* 43 (7) (2020) zsz306.
- [48] I. Choi, W. Sung, Single-channel ecg-based sleep stage classification with end-to-end trainable deep neural networks, in: 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2023, pp. 1–4.
- [49] F. Cai, M.M.R. Siddiquee, T. Wu, V.M. Lubecke, O. Borić-Lubecke, TP-CL: a novel temporal proximity contrastive learning approach for obstructive sleep apnea detection using single-lead electrocardiograms, *Biomed. Signal Process. Control* 100 (2025) 106993.
- [50] M. Tang, Z. Zhang, Z. He, W. Li, X. Mou, L. Du, P. Wang, Z. Zhao, X. Chen, X. Li, et al., Deep adaptation network for subject-specific sleep stage classification based on a single-lead ECG, *Biomed. Signal Process. Control* 75 (2022) 103548.
- [51] C.-E. Kuo, G.-T. Chen, P.-Y. Liao, An EEG spectrogram-based automatic sleep stage scoring method via data augmentation, ensemble convolution neural network, and expert knowledge, *Biomed. Signal Process. Control* 70 (2021) 102981.
- [52] C.-E. Kuo, P.-Y. Liao, Y.-S. Lin, A self-attention-based ensemble convolution neural network approach for sleep stage classification with merged spectrogram, in: 2021 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2021, pp. 1262–1268.
- [53] N.E. Huang, Hilbert-Huang transform and its applications, 16, World scientific, 2014.
- [54] M. Lin, J. Wu, J. Meng, W. Wang, J. Wu, Screening of retired batteries with gramian angular difference fields and ConvNeXt, *Eng. Appl. Artif. Intell.* 123 (2023) 106397.
- [55] S. Cui, G. Yang, J. Guan, Y. He, X. Zhou, M. Bi, H. Shen, Y. Xu, A cuffless blood pressure estimation method using dimensionality increasing and two-dimensional convolution, *IEEE J. Biomed. Health Inform.* (2025).
- [56] T. Lei, J. Li, K. Yang, Time and frequency-domain feature fusion network for multivariate time series classification, *Expert Syst. Appl.* 252 (2024) 124155.
- [57] T. Ridnik, E. Ben-Baruch, N. Zamir, A. Noy, I. Friedmann, M. Protter, L. Zelnik-Manor, Asymmetric loss for multi-label classification, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 82–91.
- [58] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [59] Z. Leng, M. Tan, C. Liu, E.D. Cubuk, X. Shi, S. Cheng, D. Anguelov, Polyloss: a polynomial expansion perspective of classification loss functions, *arXiv preprint arXiv:2204.12511*. (2022).
- [60] S.F. Quan, B.V. Howard, C. Iber, J.P. Kiley, F.J. Nieto, G.T. O'Connor, D.M. Rapoport, S. Redline, J. Robbins, J.M. Samet, et al., The sleep heart health study: design, rationale, and methods, *Sleep* 20 (12) (1997) 1077–1085.
- [61] A.L. Goldberger, L.A.N. Amaral, L. Glass, J.M. Hausdorff, P.C. Ivanov, R.G. Mark, J.E. Mietus, G.B. Moody, C.-K. Peng, H.E. Stanley, PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals, *Circulation* 101 (23) (2000) e215–e220.
- [62] Y. Zhao, Y. Ran, X. An, Y. Zheng, A spatiotemporal attention multitasking sleep staging method based on ECG signal, in: *International Conference on Algorithms, High Performance Computing, and Artificial Intelligence (AHPCAI 2024)*, 13403, SPIE, 2024, pp. 354–362.
- [63] E. Urtnasan, J.-U. Park, E.Y. Joo, K.-J. Lee, Deep convolutional recurrent model for automatic scoring sleep stages based on single-lead ECG signal, *Diagnostics* 12 (5) (2022) 1235.
- [64] B.M. Mathunjwa, Y.-T. Lin, C.-H. Lin, M.F. Abbod, M. Sadrawi, J.-S. Shieh, Automatic IHR-based sleep stage detection using features of residual neural network, *Biomed. Signal Process. Control* 85 (2023) 105070.
- [65] J. Lyu, W. Shi, C. Zhang, C.-H. Yeh, A novel sleep staging method based on EEG and ECG multimodal features combination, *IEEE Trans. Neural Syst. Rehabil. Eng.* 31 (2023) 4073–4084.
- [66] S. Davidson, C. Roman, J. Carter, M. Harford, L. Tarassenko, Sleep staging using wearables and deep neural networks, in: 2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI), IEEE, 2023, pp. 1–4.
- [67] M. Bahrami, M. Forouzanfar, Detection of sleep apnea from single-lead ECG: comparison of deep learning algorithms, in: 2021 IEEE International Symposium on Medical Measurements and Applications (MeMeA), IEEE, 2021, pp. 1–5.
- [68] M. Bahrami, M. Forouzanfar, Sleep apnea detection from single-lead ECG: a comprehensive analysis of machine learning and deep learning algorithms, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–11.
- [69] S. Hu, W. Cai, T. Gao, M. Wang, A hybrid transformer model for obstructive sleep apnea detection based on self-attention mechanism using single-lead ECG, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–11.
- [70] H. Liu, S. Cui, X. Zhao, F. Cong, Detection of obstructive sleep apnea from single-channel ECG signals using a CNN-transformer architecture, *Biomed. Signal Process. Control* 82 (2023) 104581.
- [71] Y. Hou, B. Wang, C. Zhang, Q. Wang, J. Li, P. Meng, Y. Zhang, C. Han, F. Hong, T. Zhang, OSASformer: A transformer-based model for OSAS screening via multi-source representation fusion, *Knowl. Base. Syst.* 316 (2025) 113365.